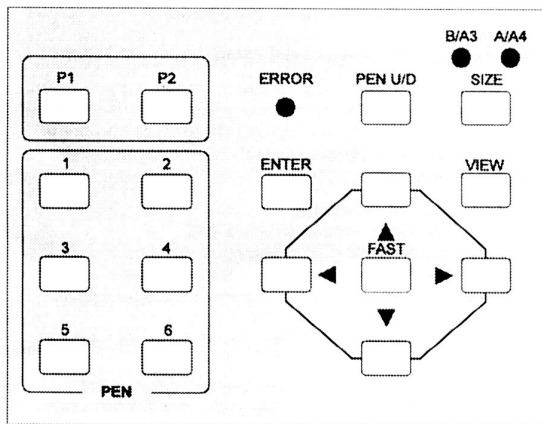


Plotting method	Multi-pen plotter
Plotting speed	Pendown 38.1 cm/s; pen up 50.8 cm/s
Resolution	Smallest addressable move 0.025mm
Paper Handling	A3- and A4-size paper and transparency film.
Interfaces	S-232C or HP Interface Bus (HP-IB) (IEEE-488)
Emulations	HP Graphics Language (HP-GL)

Control Panel



ERROR light: Indicates plotter error condition

B/A3, A/A4 lights: indicate current selected paper size.

P1, P2 keys: On power-up, raises pen and moves it to default position (lower left) of A/A4 paper size, or raises pen and moves it to default position (lower left) of B/A3 paper size. When either one is pressed together with **Enter** key, will establish new location of scaling point **P1** or **P2**.

PEN U/D key: Reverses the current pen state (up or down).

SIZE key: When pushed simultaneously with **Enter** key, selects paper size as indicated by size lights.

PEN keys: Causes plotter to retrieve same pen number from carousel.

ENTER key: Multi-use key for changing paper size and location of scaling points **P1** and **P2**.

Cursor keys: Move pen in the direction of the arrow. Using adjacent keys will move the pen at a 45 degree angle.

FAST key: When used with any cursor key, will increase pen speed 4X.

VIEW key: Turns Error light on, suspends the pen plotting, raises pen to manually change pen and view paper. When pressed again, error light will turn off, return to last coordinates and up or down status, and resume printing.

Self-test

Basic test:

1. Make sure six pens are installed in carousel and plotter is on.
2. With paper loaded, lower PAPER LOAD lever to PAPER HOLD position.
3. Press a **PEN** key to select a pen and then use **Cursor** keys to test selected pen.

Demonstration plot - Perform the following procedure to draw a bar, pie and line chart.

1. Make sure six pens are installed in carousel and plotter is off.
2. With A/M-size paper loaded, lower PAPER LOAD lever to PAPER HOLD position.
3. While holding the **P1** and **P2** keys, turn plotter on.

Troubleshooting test - The following procedure exercises both motor drive circuits, motors and encoders, the servo chips, error light circuit, gate arrays, microprocessor and ROM.

1. Make sure paper is loaded and plotter is on.
2. Manually move pen carriage near center of its travel.
3. Hold **ENTER** key while turning plotter on. The **ERROR** light should remain on.
4. Press <-- key.
5. If successful, the **ERROR** light should turn on and off continuously, and pen carriage and paper should move left and right about 6.4mm continuously.
6. Press **ENTER** key to pause test and <-- key to resume test. To terminate test, turn plotter off and back on.

Plotter Configuration

The plotter's interface is configured through a bank of DIP switches located on the rear panel. The DIP switches will vary according to which interface (RS-232C or HP-IB) is installed.

DIP Switches - RS-232C

Switch Label - Description

A3/A4 - Selects B/A3-size or A/M-size paper.

BAUD RATE

S1/PARITY - Toggles PARITY on and off.

S2/PARITY - If switch S1/PARITY is set to 1 (switch open), selects odd or even parity.

MET/US - Selects maximum plotting area. If B/A3-size paper is selected, MET selects 275x402mm and US 1 0.2x16.3 in. If A/M-size paper is selected, MET selects 192x275mm and US 7.5x1 0.2 in.

D/Y - In the Y position, received data is retransmitted and plotter does not respond unless it receives a "Plotter On" command. In the D position, plotter responds to all commands.

DIP Switches - HP-IB

Switch Label - Description

ADDRESS - Five of the seven DIP switches set the HP-IB address in binary coded decimal.

A3/A4 - Selects B/A3-size or A/A4-size paper.

MET/US - Selects maximum plotting area. If B/A3-size paper is selected, MET selects 275x402mm and US 1 0.2x16.3 in. If A/M-size paper is selected, MET selects 192x275mm and US 7.5x10.2 in.

Baud Rate Selection * - RS-232C

	One Stop Bit				Two Stop Bits			
Baud Rate	B4	B3	B2	B1	B4	B3	B2	B1
External	-	-	-	-	0	0	0	0
75	-	-	-	-	0	0	0	1
110	-	-	-	-	0	0	1	0
150	0	0	1	1	-	-	-	-
200	0	1	0	0	-	-	-	-
300	0	1	0	1	1	0	1	1
600	0	1	1	0	1	1	0	0
1200	0	1	1	1	1	1	0	1
2400	1	0	0	0	1	1	1	0
4800	1	0	0	1	1	1	1	1
9600	1	0	1	0	-	-	-	-

*1 = switch open; 0 = switch closed.

Common Problems and Fixes

Plotter does not respond to control panel, ERROR lights is off, and the PAPER lever is in the load position:

1. Check rear-panel line fuse, voltages and power supply fuses on PCA.
2. Check 4 MHz clock and Gate Array B.

Plotter responds to control panel but not to host:

1. Make sure interface connection is properly seated at both ends.
2. Test the I/O circuits by sending "SP1;SP2" from host.

Pen up/down does not work:

1. Check fuse A1F2 and pen supply voltage.
2. Check solenoid continuity.

Diagonal lines are not straight:

1. Check for defective pen drive motor/encoder assembly (especially if diagonals askew near horizontal lines).
2. Check for defective paper drive motor/encoder assembly (especially if diagonals askew near vertical lines).
3. Check for deposits on grit wheels, pinchrollers and slider rod. Clean slider rod and pinchrollers with a dry wipe.

Critical Adjustments

Note: Before performing any of the following procedures, remove top cover (see "Cover Removal" below).

Adjusting pen height - Perform the following procedure if pen carriage assembly is disassembled or replaced.

1. Position pen holder at center of platen.
2. Use a 100mm ruler to measure distance from platen to bottom of pen holder. It should be 10.5mm.
3. Insert a 0.050 inch Allen wrench through hole at rear of pen carriage. Turn clockwise to decrease pen height or counter-clockwise to increase pen height.

Removing pen carousel housing:

1. Disconnect pen carousel cable from J8 on main printed circuit board.
2. Remove screw that secures pen carousel housing to chassis assembly.
3. Tip pen carousel housing forward and lift it straight up.

Removing paper drive motor assembly:

1. Disconnect paper drive motor cable (twisted pair) from J3 and flat encoder cable from J1 on main printed circuit assembly.
2. Remove pen drop shield.
3. Loosen motor clamp and remove motor from its mounting.

Removing pen solenoid:

1. Loosen paper drive motor clamp enough to slide motor right about 3/4 inch. Lift right end of motor slightly to release motor from chassis.
2. Disconnect solenoid cable from J2 on printed circuit assembly.
3. Loosen solenoid mounting screw enough to allow solenoid removal.

Removing pen drive motor and belt:

1. Disconnect pen drive motor cable (twisted pair) from J5 and flat encoder cable from J6 on main printed circuit assembly.
2. Remove belt tensioner by pressing down on tensioner and sliding the tang at bottom out of chassis slot.
3. Loosen pen drive motor clamp and remove motor.
4. Slide belt from pen carriage to remove it.
5. Loosen pen solenoid mounting screw and slide solenoid to the right. Remove armature and spring.
6. Slide pen lift bar just far enough right to allow belt removal.

Removing pen carriage, pen holder, and damper:

1. Remove paper drive motor (see above).
2. Remove pen solenoid (see above).
3. Remove pen drive motor (see above).
4. Slide belt from pen carriage and move carriage to the left.
5. Remove end bearing cap while sliding pen lift bar to the right and out of carriage assembly.
6. Move slider rod to the right just far enough to release left end of rod from its mounting. Slide rod to the left and out of carriage/pen holder assembly.
7. Carefully remove plastic damper from carriage and pen holder.

Note: After reassembling the above, measure the pen down force with a gram gauge before replacing the top cover. Turn plotter on. Lower a pen onto platen. Place tip of gauge under lip of pen body and make sure that pen just starts to lift with 19 +/- 10 grams. If it does not, replace the pen down spring.

Removing printed circuit assembly:

1. Disconnect all cables from printed circuit board.
2. Remove pen carousel housing (see above).
3. Remove screw holding control panel and remove panel.
4. Remove recessed screw between pen solenoid and pen drive motor.
5. Lift right side of chassis assembly. Tabs on left side will release from base plate.
6. Remove screws or studs holding rear panel interface connector.
7. Lift front end of printed circuit assembly and remove it from base plate.

Cover Removal

1. Turn plotter off and disconnect power cord and interface cable.
2. Remove 3 screws at rear of plotter.
3. Lift rear of top cover so that the front releases from base.

Field Replaceable Units

DESCRIPTION	OEM P/N
Damper, silicone rubber	07475-4000
Paper drive motor assembly	07470-6017
PCA-Main HP-IB	07475-6010
PCA-Main RS-232C	0747506010
Pen carriage	5040-8650
Pen drive motor assembly	07470-6018
Pen carousel motor	3140-0687
Pen carousel assembly	5061-5080
Pen holder	07475-6002
Solenoid	07475-6001
Spring, pen down	1460-1950

Note: Be sure to print a menu/configuration list before replacing a PCS or logic containing configuration settings.

IBM P/N: 55X3584
IBM machine type: 1538-B01, 1516-H20

Tech Support 800-877-7764

HP 7475A Color Desktop Plotter



Technical Data

**The *one* plotter for
both CAD and color
business graphics**

Features

- Two small-format media sizes
- Full range of pen/media combinations
- High-quality output
- Strong software support
- Hardware compatibility
- Built-in intelligence

The HP 7475A color desktop plotter produces high-quality A4/A- and A3/B-size color graphics for business and PC CAD applications. The HP 7475A is ideal for professionals who need the larger drawings for PERT charts, flow charts, project schedules, and design applications. It also produces professional-quality color overheads for presentations and colorful summary charts for handouts and reports.

Full Range of Pen/Media Combinations

HP 7475A users can select from paper (regular and glossy), overhead transparency film, and durable double-matte polyester film. Fiber-tip pens for paper and transparencies come in 10 bright colors and two tip widths. Refillable liquid-ink pens are available for final-quality drawings on polyester film.

High-quality Output

The HP 7475A color desktop plotter combines high resolution and excellent repeatability to ensure professional-quality output. It has an addressable resolution of 0.025 mm (0.001 in.), so it can plot up to 1000 points in a one-inch line.

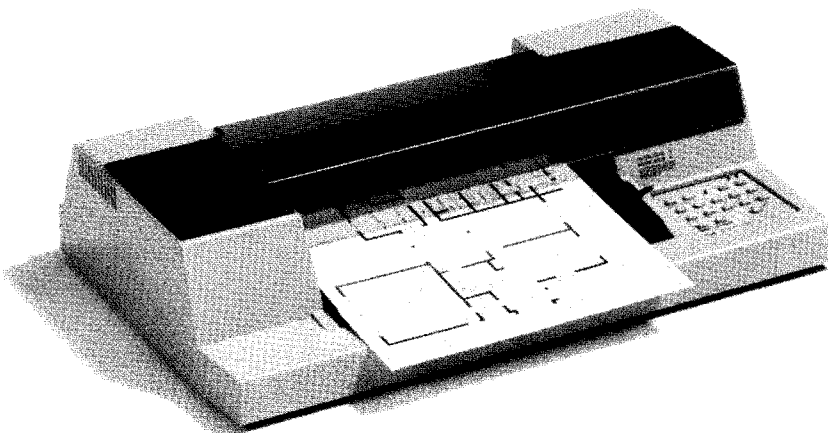
When commanded to return to the same point with no pen change, it does so within 0.1 mm (0.004 in.). This precision means your drawings have straight lines, crisp characters, circles that close, and corners that meet.

Strong Software Support

One of the most widely-supported plotters in the world, the HP 7475A works with major software packages from the entire spectrum of graphics applications—integrated software, business graphics/presentations, computer-aided-design/drafting, scientific and statistical analysis, project scheduling, and more.

Hardware Compatibility

A choice of two interfaces makes it easy to add an HP 7475A to almost any system—personal computers, mainframes, even smart instruments. The HP 7475A comes with RS-232-C/CCITT V.24 and HP-IB (IEEE 488-1978) interface options. And with an additional eavesdrop cable, the RS-232-C plotter can be connected in series between a computer and a terminal.



Built-in Intelligence

The HP 7475A's control-panel keys can alter the size of the drawing area when switching media sizes or rotate the plot 90 degrees. Control-panel keys can also be used in conjunction with a digitizing sight.

Programmers will appreciate the intelligence features which eliminate the need for software-generated characters and functions. Hewlett-Packard Graphics Language (HP-GL) instructions (more than 50) govern such tasks as labeling, pen movement, and drawing arcs and circles.

Technical Information

Media Sizes

210 × 297 mm (ISO A4)

8½ × 11 in. (ANSI A)

297 × 420 mm (ISO A3)

11 × 17 in. (ANSI B)

Pens

Number: 6 in carousel

Type: Fiber-tip (paper and transparency), refillable short-body liquid-ink

Media

Paper (regular and glossy), transparency film, double-matte polyester film

Character Sets

French/German, HP 9825, Scandinavian, Spanish/Latin American, Roman Extensions, and these ISO registered sets: ANSI ASCII (006), French (025), German (021), International Reference Version (002), Italian (015), JIS ASCII (014), Katakana (013), Norwegian I (060), Norwegian II (016), Portuguese (016), Spanish (017), Swedish (010), Swedish for Names (011), United Kingdom (004)

Resolution

Addressable: 0.025 mm
(0.001 in.)

Mechanical: 0.025 mm
(0.001 in.)

Repeatability

With a given pen: 0.1 mm
(0.004 in.)

Pen to pen: 0.2 mm (0.008 in.)

Pen Velocity

Pen down, maximum: 38.1 cm/s
(15.0 in./s)

Pen up: 50.8 cm/s (20.0 in./s)

Programmable: approximately
0.4 to 38.1 cm/s (0.2 to 15.0 in./s)
in increments of 0.4 cm/s

Acceleration

Approximately 2 g

Maximum Plotting Area

Pen axis: 258.0 mm (10.2 in.)
for A/B; 275.0 mm (10.8 in.) for
A4/A3

Paper axis: 198.0 mm (7.80 in.)
for A; 192.0 mm (7.56 in.) for
A4; 414.0 mm (16.3 in.) for B;
402.0 mm (15.8 in.) for A3

Buffer Size

1024 bytes

Environmental Ranges

Temperature

Operating: 0 to 55 degrees C
(32 to 131 degrees F)

Storage: -40 to 75 degrees C
(-40 to 167 degrees F)

Humidity

Operating: 5 to 95% (at 40
degrees C)

Power Requirements

Source: 100, 120, 220, 240 V,
-10%, +5%

Frequency: 48-66 Hz

Consumption: 35 W maximum

Physical Specifications

Height: 127.0 mm (5.0 in.)

Width: 568.0 mm (22.4 in.)

Depth: 367.0 mm (14.5 in.)

Net weight: 7.0 kg (16.0 lb)

Shipping weight: approx.
11.0 kg (25.0 lb)

Product Certifications

FCC certified to conform to
limits set for radio frequency
when used with a Class B com-
puting device; meets or exceeds
IEC-380, IEC-435, IEC-478, and
CSA C22.2 No. 154.

Interface and Cable Requirements

System	Cable
HP Vectra PC with HP 24540A/B serial/parallel interface card or HP 24541A/B dual serial interface card using 9-pin connector	HP 24542G
HP Vectra PC with HP 24541A/B dual serial interface card using 25-pin connector	HP 17255M
HP Touchscreen PC using serial interface	HP 17255M
HP Touchscreen PC using HP-IB interface	HP 10833A (1.0 m) HP 10833B (2.0 m) HP 10833C (3.0 m) HP 10833D (0.5 m)
HP 3000 or DEC VAX in eavesdrop configuration	HP 17455A
IBM PS/2, PC, PC-XT, and compatibles	HP 17255D
IBM AT and compatibles	HP 24542G
Apple Macintosh 128K and 512K	HP 92219M
Apple Macintosh SE, Macintosh Plus, and Macintosh II	HP 17302A
Apple IIe	HP 17355M

Ordering Information

Standard Unit

7475A

Color desktop plotter

Option numbers:

001 RS-232-C/CCITT V.24 interface

002 HP-IB (IEEE 488-1978) interface

Note: Interface cables must be ordered separately

Accessories Included
Operation and Interconnection Manual (language appropriate to plotter destination)

Reference Card

Power cord (appropriate to plotter destination)

Plotter paper
50 sheets, A4/A size (paper sizes determined by plotter destination)

Transparency film sampler, A4/A size (film sizes determined by plotter destination)

Six-pen carousel for fiber-tip pens

Assorted fiber-tip paper and transparency pens

HP Plotter Supplies Catalog

Accessories Available

07475-90000	Hardware Support Manual
07475-90001	Interfacing/Programming Manual
5061-5080	Additional fiber-tip pen carousel
07475-60030	Liquid-ink drafting pen carousel

For a complete list of available supplies, check the current version of the HP Plotter Supplies Catalog or the Hewlett-Packard PC Peripherals Price Guide.

United States:

Hewlett-Packard Company
4 Choke Cherry Road
Rockville, MD 20850
301 670 4300

Hewlett-Packard Company
5201 Tollview Drive
Rolling Meadows, IL 60008
312 255 9800

Hewlett-Packard Company
5161 Lankershim Blvd.
No. Hollywood, CA 91601
818 505 5600

Hewlett Packard Company
2015 South Park Place
Atlanta, GA 30339
404 955 1500

Canada:

Hewlett-Packard Ltd.
6877 Goreway Drive
Mississauga, Ontario L4V1M8
416 678 9430

Australia/New Zealand:

Hewlett-Packard Australia Ltd.
31-41 Joseph Street
Blackburn, Victoria 3130
Melbourne, Australia
03 895 2895

Europe/Africa/Middle East:

Hewlett-Packard S.A.
Central Mailing Department
P.O. Box 529
1180 AM Amstelveen
The Netherlands
31 20/547 9999

Far East:

Hewlett-Packard Asia Ltd.
22/F Bond Centre
West Tower
89 Queensway
Central, Hong Kong
5 848 7777

Japan:

Yokogawa Hewlett-Packard Ltd.
29-21, Takaido-Higashi 3-chome
Suginami-ku, Tokyo 168
03 331 6111

Latin America:

Latin American Region Headquarters
Monte Pelvoux Nbr. 111
Lomas de Chapultepec
11000 Mexico D.F., Mexico
905 596 79 33

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COMPUTER PERIPHERALS

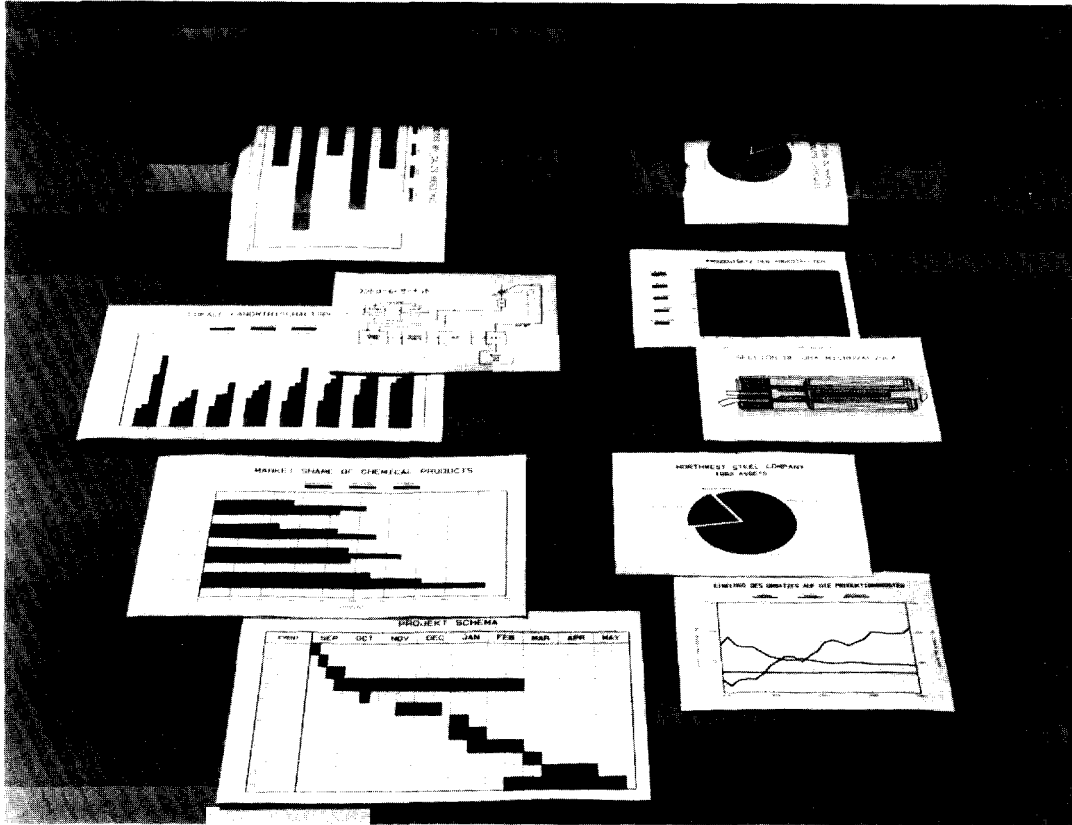
Graphics Plotters
Models 7470A and 7475A

99



- Low cost, high performance
- Choice of six- or two-pen models

- Plot on paper
- Plot on HP overhead transparency film



Hewlett-Packard's HP 7470A and 7475A graphics plotters provide the kind of graphics excellence you would expect to find only in much more expensive plotters. They feature the same high-quality components and innovative paper-moving technology which were introduced in HP's drafting plotters and which make it possible to offer high performance plotters at affordable prices. Refer to the table below for a quick comparison of the main features available in each plotter.

Features	HP 7475A	HP 7470A
Media sizes	Two ANSI sizes: A (8.5 x 11 in.) and B (11 x 17 in.) Two ISO sizes: A4 (210 x 297 mm) and A3 (297 x 420 mm)	One ANSI size: A (8.5 x 11 in.) One ISO size: A4 (210 x 297 mm)
Pens	Six fiber-tip; programmable pen selection; automatic capping	Two fiber-tip; programmable pen selection; automatic capping
HP-GL Instructions	More than 50 instructions	More than 40 instructions
Character sets	19 sets, including ISO European standards and Katakana	Five sets
Standard Interfaces	HP-IB (IEEE 488-1978) or RS-232-C (CCITT V.24)	HP-IB (IEEE 488-1978) or RS-232-C (CCITT V.24) or HP-IL
Technology	Each plotter incorporates one permanent interface option.	
	Both plotters use the same micro-grip drive for paper movement and have the same high resolution, repeatability, and velocity.	

Computer Applications

The HP 7470A and 7475A provide hardcopy computer graphics for technical, scientific, and business applications. Colorful A4/A-size charts and graphs are ideal for reports and overhead transparencies. Use them for summarizing data, identifying trends, comparing results, and focusing on exceptions. The larger A3/B-size plots that can be drawn on the HP 7475A are particularly useful for time lines, PERT charts, schematics, engineering drawings, and other applications where you need to show visual detail.

Measurement Applications

The HP 7470A and 7475A add hardcopy graphics capability to intelligent instruments and instrument systems with HP-IB (IEEE 488-1978). For most applications that use a display screen and an oscilloscope camera, these plotters can produce high-quality hardcopy of the screen for a cost that is substantially lower than camera film. Because they plot directly from measured data, they eliminate problems created by distortion from the screen. And plotter output provides better visual resolution than photographs. Many systems without screen displays can also have the benefits of HP 7470A or 7475A hardcopy graphics at very little additional cost.

Easy to Use

When the HP 7475A or 7470A plotters are turned on, default conditions are automatically established for most plotting parameters. In many cases, it is only necessary to load the pens and plotting medium in order to start plotting.

Media and pen loading are also easy. A guide control lever makes media alignment perfect every time. The front panel can be used to select pens, to halt the program for exchanging pen colors, or to move the plot forward to "view" what you have plotted.

The front panel also allows easy access to the plotter's digitizing capability and scaling points. And, on the HP 7475A, push buttons can rotate plots 90 degrees or run a demonstration plot directly from the plotter.

Intelligence Features

Intelligence features are built directly into these plotters to save you time by eliminating the need for software-generated characters and functions. Many HP-GL instructions (more than 50 in the HP 7475A; more than 40 in the HP 7470A) govern such tasks as labeling, pen movement, drawing arcs and circles, and selecting from a large variety of character sets. The HP 7475A has 19 character sets in-



COMPUTER PERIPHERALS

Graphics Plotters

Model 7470A and 7475A (cont'd)

cluding ISO European sets, Katakana, ASCII, and Roman 8 extensions; the HP 7470A has five internal character sets.

The HP 7475A's extra HP-GL instructions, which are used for filling rectangles and wedges for pie and bar charts, provide an enhancement especially designed for professional graphics.

Writing Systems

The HP 7470A has two built-in pen stalls which make two-color plotting easy. For plots with more than two colors, the program can be halted through program or front panel control; new pens can then be installed and plotting resumed. The HP 7475A's six-pen carousel allows you to store up to six different pen colors or a variety of colors and widths.

Several automatic features are included to protect the tip of the pen and increase pen life. When housed in the stall or carousel, the pen is capped to prevent premature drying. When a pendown command is given, the pen force is damped and the pen is gently lowered to the plotting surface.

High-Quality Output

The HP 7470A and 7475A have an addressable step size of 0.025

mm (0.001 in.). With this resolution, they can plot up to 1000 points in a 1-inch line. When commanded to return to the same point with no pen change, they achieve this repeatability within 0.1 mm (0.004 in.) Because of this outstanding resolution and repeatability, both plotters produce straight lines and smooth circles that have an artist-drawn appearance.

Interface Options

The HP 7475A and 7470A are easy to interface with most HP and non-HP computers. Both plotters offer the RS-232-C/CCITT V.24 or HP-IB (IEEE 488-1978) interface. With the RS-232-C option, a dual input/output cable is available for connecting the plotters with a terminal and computer. In addition, the HP 7470A offers a third interface option, HP-IL. This interface is used to connect the plotter with low-cost, portable HP systems.

Graphics Software

HP offers a full line of graphics software packages for use on most HP computer products. And software is also available for many non-HP computers. These packages make it easy for non-programmers to use the HP 7470A and 7475A plotters. Details are available from any HP sales and support office.

Specifications

	HP 7475A	HP 7470A
Resolution	Smallest addressable step size: 0.025 mm (0.001 in.)	
Repeatability	With a given pen: 0.1 mm (0.004 in.) From pen to pen: 0.2 mm (0.008 in.)	
Pen velocity (each axis)	Pen up, 50.8 cm/s (20 in./s); pen down, maximum — 38.1 cm/s (15 in./s), programmable — 1 to 38 cm/s in 1 cm/s increments	
Acceleration	Approximately 2 g's	
Environmental range	Operating, 0°C to 55°C Non-operating, -40°C to 75°C	
Plotting area		
X-axis	258 mm (10.2 in.), A/B 275 mm (10.8 in.), A4/A3 198 mm (7.80 in.), A 192 mm (7.56 in.), A4 414 mm (16.3 in.), B 402 mm (15.8 in.), A3	191 mm (7.5 in.) A 191 mm (7.5 in.), A4 257 mm (10.2 in.), A 272 mm (10.7 in.), A4
Y-axis		
Interfaces	HP-IB (IEEE 488-1978), implements the following HP-IB functions as defined in IEEE 488-1978: SH1, AH1, T2, TE0, LE0, SR1, RL0, DC1, DT0, L2, PP0. (listen only or address less than 7, otherwise PP2)	
	RS-232-C/CCITT, asynchronous serial ASCII with switch selectable baud rates of 75, 110, 150, 200, 300, 600, 1200, 2400, 4800, 9600. External clock input capabilities with intermediate baud rates of up to 9600 baud. 1024 byte buffer.	Same as HP 7475A except 255 byte buffer. HP-IL, Hewlett-Packard Interface Loop for use with portable systems.
Power Requirements	Source: 100, 120, 200, 240 V~ -10%, +5% Frequency: 48-66 Hz	
	Consumption: 35 W maximum	Consumption: 25 W maximum
Size:		
Height	127 mm (5 in.)	127 mm (5 in.)
Width	568 mm (22.4 in.)	432 mm (17 in.)
Depth	367 mm (14.5 in.)	343 mm (13.5 in.)
Weight:		
Net	7 kg (16.0 lb)	6 kg (13.5 lb)
Shipping	Approx. 11 kg (25.0 lb)	Approx. 10 kg (22.0 lb)
FCC	FCC certified to conform to limits set for radio frequency interference when used with a Class B computing device.	

Accessories Supplied

HP 7475A

HP 07475-90001 Interfacing and Programming Manual
HP 07475-90002 Operation and Interconnection Manual
HP 07475-90004 Reference Card

HP 7470A

HP 07470-90001 Interfacing and Programming Manual
HP 07470-90002 Operator's Manual
HP 07470-90003 Interconnection Guide
HP 07470-90004 Reference Card

Power cords and an assortment of pens and drawing media are also supplied with the plotters. The media size and the appropriate power cord are determined by plotter destination. The HP-IL cable (½-metre) is supplied with Option 003 only.

Note: Interface cables are not supplied with Option 001 and Option 002 plotters.

Ordering Information

Options

001 RS-232-C/CCITT V.24 (cable not included)

Price

N/C

002 HP-IB (IEEE 488-1978) (cable not included)

N/C

003 HP-IL for 7470A only (cable included)

N/C

Note: Option 001, 002, or 003 must be specified when ordering HP 7470A; Option 001 or 002 must be specified with HP 7475A.

Interface Cables

HP 13242G Male-male, special RS-232-C cable for use

\$69

with Option 001, HP 150 Personal Computer

HP 17255D Male-female, special RS-232-C cable for

\$50

use with Option 001, IBM Personal Computers

HP 17355F Male-male standard cable for use with Option 001

\$52.50

HP 17455A Eavesdrop cable for use with Option 001

\$75

HP 10833A or HP 45529A or HP 31389A HP-IB

\$81

1-metre cable for use with Option 002

HP 82167A HP-IL ½-metre cable (included with Option 003)

\$6

Plotters

HP 7470A Two-pen Graphics Plotter

\$1095

HP 7475A Six-pen Graphics Plotter

\$1895

Interface Cables for HP Graphics Plotters and Personal Printers

TECHNICAL DATA, MARCH 1985



Most HP plotters are not shipped with an interface cable. Hewlett-Packard graphics plotters (HP 7470A, 7475A, 7550A), drafting plotters (HP 7580B, 7585B, 7586B), and personal printers (HP Laserjet, HP Thinkjet) can be used with a variety of HP and non-HP personal computers and computer systems. However, because different configurations often require different interface cables, customers must order the appropriate cable separately.

To find the proper cable to connect your plotter or printer to your personal computer, check the table below. To determine the proper cable for other configurations (such as connecting your plotter or printer to a mainframe computer

terminal), consult the tables on Page 2. Cable schematics appear on Page 3. For ordering information, see Page 4.

Standalone Configurations:

In a standalone configuration, the plotter/printer is connected to its own dedicated computer port. This setup is primarily for personal computers in workstation environments, but also includes peripherals on their own computer port in mainframe environments.

Note: Cables from non-HP computer manufacturers are listed for information purposes only. Check with the manufacturer for current model numbers and prices.

Printer/Plotter to Personal Computer, Standalone (RS-232-C)			
Computer System	HP Laserjet HP Thinkjet	HP 7470A, 7475A, 7580B, 7585B, 7586B	HP 7550A
HP Series 150, TI Pro, Wang PC	HP 13242G	HP 13242G	HP 17255D
The HP PORTABLE	HP 92221P	HP 92221P	HP 92221P and HP 17355F*
IBM PC and PC/XT, Compaq, AT&T PC 6300	HP 17255D	HP 17255D	HP 17255F
IBM AT	HP 17255D and IBM 6450242 Serial Adapter Cable	HP 17255D and IBM 6450242 Serial Adapter Cable	HP 17255F and IBM 6450242 Serial Adapter Cable
IBM PCjr	HP 17255D and IBM Serial Device Adapter Cable	HP 17255D and IBM Serial Device Adapter Cable	HP 17255F and IBM Serial Device Adapter Cable
Apple IIe, DEC Rainbow, Osborne	HP 17355M	HP 17355M	HP 17355D
Apple III	HP 17355M** and Apple A3M0019	HP 17355M** and Apple A3M0019	HP 17355D and Apple A3M0019
Apple IIc	Apple A9C0308	Apple A9C0308	HP 17355F* and Apple A9C0308

*If the combination of these two cables (over 4 m) is too long, order HP 92222F instead of HP 17355F; second cable is still necessary.

**If the combination of these two cables (over 3 m) is too long, order HP 92222M instead of HP 17355M; the Apple A3M0019 is still necessary.

Printer/Plotter to Personal Computer (HP-IB/IEEE-488)		
Computer System	HP Thinkjet	HP 7470A, 7475A, 7550A, 7580B, 7585B, 7586B
HP Series 80, HP Series 150, HP 9000 Series 200, 500		HP 45529A/B or HP 10833A/B or HP 31389A/B

Printer to Mainframe, Standalone (RS-232-C)	
Computer System	HP Laserjet only
HP 3000 Series 37 ATP	HP 92218D

Plotter to Mainframe, Standalone (RS-422-A)	
Computer System	HP 7550A only
HP 3000 Series 64, 39, 42, 48	HP 17855A

Plotter to Calculator/Handheld Computer (HP-IL)	
Computer System	HP 7470A only
HP Series 40 Calculator HP Series 70 Handheld Computer	HP 82167A

Eavesdrop Configurations:

In an eavesdrop configuration, the plotter is situated between a mainframe computer (i.e., HP 3000, IBM 360/370, DEC VAX) and a terminal or a personal computer being used as a terminal. Both the terminal and plotter share the same RS-232-C port from the mainframe computer.

To use an HP 7550A, 7580B, 7585B, or 7586B in eavesdrop mode, the only additional cable required in most configura-

tions is the one from the plotter to the mainframe computer. These plotters have two RS-232-C ports, so the terminal to mainframe computer cable is used with the second RS-232-C port on the plotter.

When connecting an HP 7470A or 7475A in eavesdrop mode, a special eavesdrop cable (HP 17455A) is required in addition to the cable from the plotter to the mainframe computer.

Eavesdrop Configuration (RS-232-C)			
Computer System	HP 7470A, 7475A	HP 7580B, 7585B, 7586B	HP 7550A
HP Series 150 (in terminal mode)	HP 17455A and HP 13242N	HP 17355M or HP 13242N	HP 17355M or HP 13242N
HP 3000	HP 17455A and cable from computer	HP 17355M or HP 13242N	Cable from computer

Cable Schematics

Connector Types

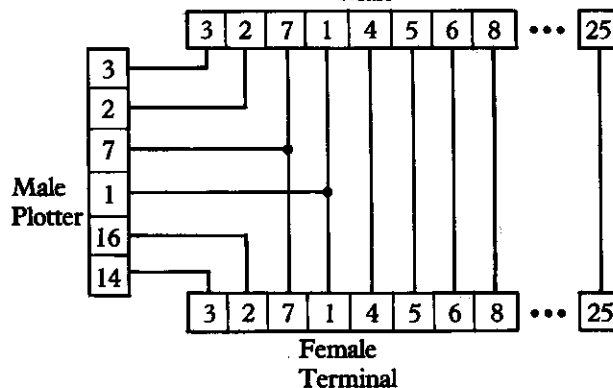
Part numbers ending in M, F, or D follow this convention:

M = Male to Male
F = Female to Female
D = Male to Female

Y-Cable for Eavesdrop Connection

HP 17455A

Computer
Male



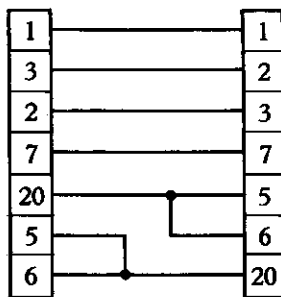
Pins 4, 5, 6, and 8 through 25 are directly connected between the computer and terminal connectors.

Modem Eliminator Type

HP 17255D/F

HP 13242G*

Length: 1 meter



Pins not shown are not used.

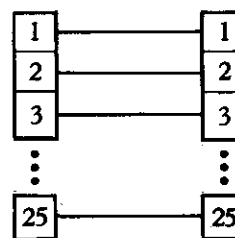
**Other pins are connected in this cable but they do not affect plotter usage. (The HP 13242G uses male-to-male connectors.)*

Straight Through

HP 17355M/F/D*

Length: 1 meter

Pin Out



Pins 1-25 are directly connected.

**Cables HP 92222F/M are of the same pin configuration as HP 17355F/M, but are only 8 inches long. (There is no HP 92222D.)*

Ordering Information

RS-232-C/CCITT V.24

Cables

HP 13242G HP 17355M
 HP 13242N HP 17455A
 HP 17255D HP 92218D
 HP 17255F HP 92221P
 HP 17355D HP 92222F
 HP 17355F HP 92222M

HP-IB (IEEE-488)

Cables

HP 10833A
 HP 10833B
 HP 31389A
 HP 31389B
 HP 45529A
 HP 45529B

RS-422-A Cable

HP 17855A

HP-IL Cable

HP 82167A

Cables are available from your local dealer or HP Sales Representative. To order cables or to obtain a copy of HP's *Computer Users Catalog* (5953-2450), which includes a complete listing of Hewlett-Packard supplies and accessories, call one of HP's Direct Order telephone numbers:

United States	800-538-8787
California	408-738-4133
United Kingdom	0734-792868
	0734-792959
France	(6) 928 32 64
Belgium/Luxembourg	(02) 762 32 00
Switzerland	(057) 31 22 54
	or 31 22 59
West Germany	07031-142829
	07031-223133
The Netherlands	020-470639
South Africa	802-5111
	53-7954
	28-4178
Canada	
Toronto Local	416-671-8383
Ontario	1-800-268-6982
Quebec	1-800-387-3417
British Columbia	112-800-387-3154
Other Provinces	1-800-387-3154
Sweden	08-7502027
	08-7502028
Australia	(03) 895-2645
	or 895-2615
	(02) 888-7712
	or 887-1611

Apple IIc, IIe, and III are products of Apple Computer, Inc. AT&T PC 6300 is a product of AT&T Information Systems, Inc. Compaq is a product of Compaq Computer Corp. DEC Rainbow and VAX are products of Digital Equipment Corp. IBM PC, XT, AT, and PCjr are products of International Business Machines Corp. Osborne is a product of Osborne Computer Corp. TI Pro is a product of Texas Instruments. Wang PC is a product of Wang Laboratories, Inc.

For more information, call your local HP sales office listed in the telephone directory white pages. Ask for a Personal Computer Representative. Or write to Hewlett-Packard: **U.S.A.** — P.O. Box 10301, Palo Alto, CA 94303-0890. **Europe** — P.O. Box 999, 1180 AZ Amstelveen, The Netherlands. **Canada** — 6877 Goreway Drive, Mississauga, L4V 1M8, Ontario. **Japan** — Yokogawa-Hewlett-Packard Ltd., 3-29-21, Takaido-Higashi, Suginami-ku, Tokyo 168. **Elsewhere** in the world, write to Hewlett-Packard Intercontinental, 3495 Deer Creek Road, Palo Alto, CA 94304.